

RealNut Intelligence (RNI)

Software Design Document (SDD)

Version: 1.0

Project Type: Internal Business Intelligence Platform

Client: Real Nut Dry Fruits

Primary Marketplace: Blinkit

Technology Stack: Python, Streamlit, SQLite/PostgreSQL

1. Executive Summary

Purpose

RealNut Intelligence (RNI) is an AI-powered Business Intelligence platform developed exclusively for Real Nut Dry Fruits' Blinkit seller operations.

The platform will consolidate sales, inventory, keyword, competition, and operational data into a single dashboard, enabling management to make faster, data-driven business decisions.

Unlike traditional reporting tools that display historical metrics, RNI focuses on providing actionable recommendations through automated analytics, predictive insights, and intelligent alerts.

2. Vision

To become the central command center for all Blinkit operations of Real Nut Dry Fruits.

The system should answer:

- What happened?
 - Why did it happen?
 - What will happen next?
 - What should we do today?
-

3. Project Objectives

Business Objectives

- Increase sales through data-driven decision making
 - Identify high-performing and underperforming products
 - Improve warehouse efficiency
 - Monitor regional demand
 - Track competitors continuously
 - Discover keyword opportunities
 - Reduce inventory losses
 - Generate automated executive reports
-

Technical Objectives

- Fully automated reporting
 - Minimal manual data processing
 - Modular architecture
 - Easily extendable analytics engine
 - AI-assisted business recommendations
 - Secure local data storage
 - Daily scheduled report generation
-

4. Scope

Included

- Blinkit sales analytics
 - Product performance analysis
 - Regional sales intelligence
 - Warehouse inventory reporting
 - Keyword opportunity analysis
 - Competitor tracking
 - Smart alerts
 - AI recommendations
 - Executive dashboard
 - PDF report generation
-

Future Scope

- Zepto integration

- Instamart integration
 - Amazon Seller integration
 - Shopify integration
 - ERP integration
 - Mobile application
 - Demand forecasting using Machine Learning
 - Automated pricing engine
-

5. Stakeholders

Role	Responsibility
Business Owner	Decision making
Operations Team	Inventory & warehouse management
Marketing Team	Promotions & keyword optimization
Management	Business strategy
Developer	System maintenance

6. Functional Modules

Module 1: Executive Dashboard

Purpose: Provide a one-page summary of overall business health.

Features:

- Total Revenue
 - Orders
 - Units Sold
 - Growth Metrics
 - Revenue Trend
 - Today's Focus
 - Smart Alerts
 - AI Recommendations
 - Product Health Overview
 - Regional Snapshot
-

Module 2: Sales Intelligence

Purpose: Analyze product and sales performance.

Features:

- Daily Sales
 - Weekly Sales
 - Monthly Sales
 - Product Rankings
 - Revenue Analysis
 - Growth Analysis
 - Product Health Score
 - Sales Trends
 - Category Analysis
-

Module 3: Regional Intelligence

Purpose: Understand geographical demand.

Features:

- Revenue by Region
 - Orders by Region
 - Top Products by Region
 - Regional Growth
 - Market Penetration
 - Regional Heat Maps
 - Warehouse Mapping
-

Module 4: Keyword Intelligence

Purpose: Improve search visibility and conversions.

Features:

- Keyword Rankings
- Search Volume
- Click Through Rate
- Conversion Rate
- Keyword Opportunity Score
- AI Listing Suggestions
- SEO Improvement Recommendations

Module 5: Competition Intelligence

Purpose: Track market competition.

Features:

- Price Comparison
 - Ratings Comparison
 - Review Comparison
 - Offer Tracking
 - Stock Availability
 - Market Position
 - AI Competitive Recommendations
-

Module 6: Warehouse Intelligence

Purpose: Optimize inventory management.

Features:

- Current Stock
 - Inventory Age
 - Daily Consumption
 - Stock Forecast
 - Overstock Alerts
 - Understock Alerts
 - Warehouse Performance
 - Restock Recommendations
-

Module 7: Smart Alert Engine

Purpose: Notify management of important business events.

Alert Categories:

- Sales Drop
- Sales Growth
- Inventory Risk
- Competitor Price Change
- Keyword Opportunity
- Review Spike
- Warehouse Issues

- High Demand Prediction

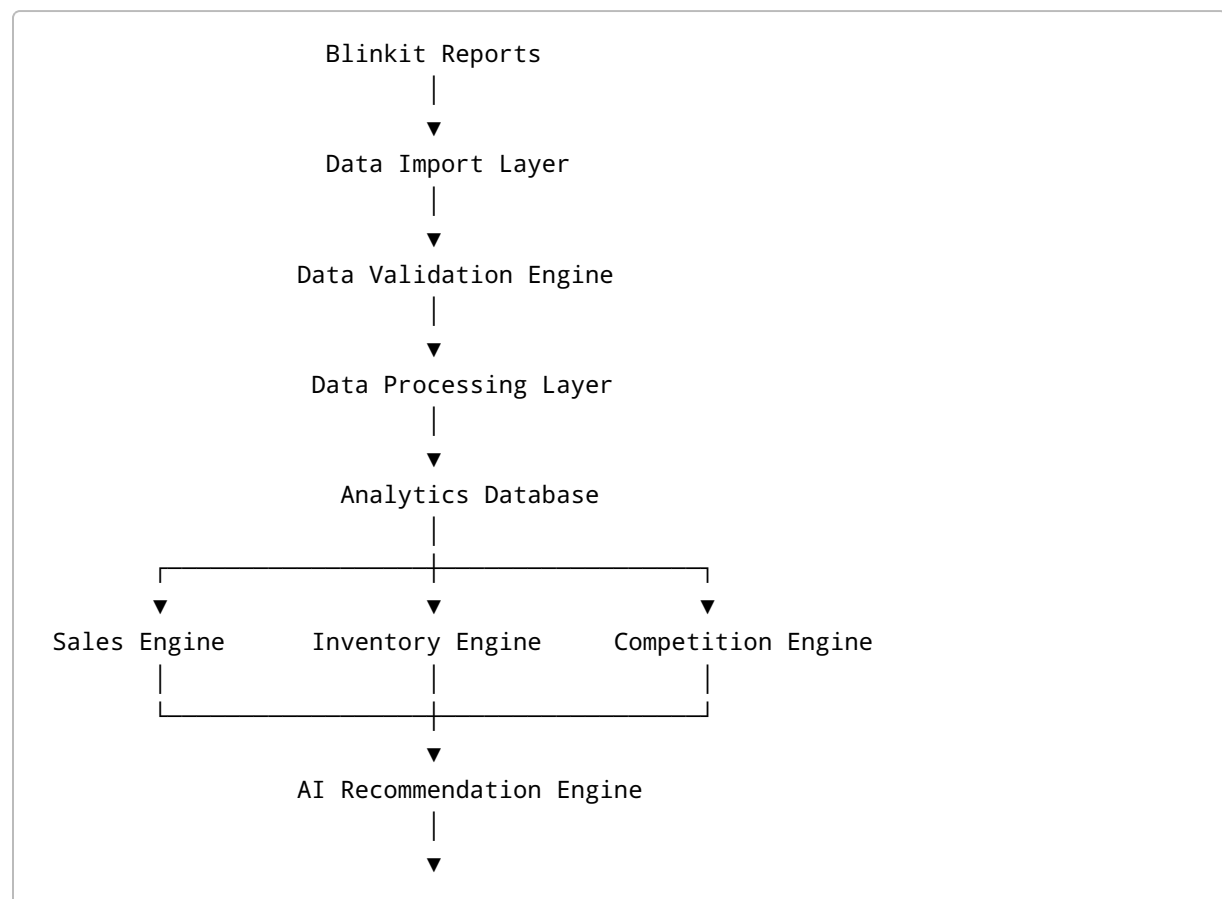
Module 8: AI Business Advisor

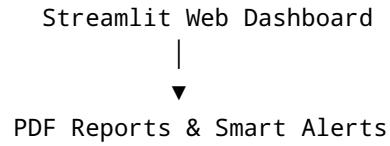
Purpose: Generate business recommendations automatically.

Capabilities:

- Daily Summary
 - Weekly Summary
 - Product Suggestions
 - Pricing Suggestions
 - Marketing Suggestions
 - Inventory Suggestions
 - Regional Expansion Suggestions
 - Competitive Strategy Suggestions
-

7. System Architecture





8. Technology Stack

Layer	Technology
Language	Python
Dashboard	Streamlit
Database	SQLite (Phase 1), PostgreSQL (Phase 2)
Data Processing	Pandas
Visualization	Plotly
Scheduling	APScheduler
AI	OpenAI API
Reporting	ReportLab
Configuration	YAML
Logging	Python Logging
Version Control	Git

9. Data Pipeline

Blinkit Export

↓

Import

↓

Validation

↓

Cleaning

↓

Normalization

↓

Database Storage

↓

Analytics

↓

AI Analysis

↓

Dashboard

↓

Executive Reports

10. Folder Structure

```
realnut_intelligence/
```

```
|
```

```
|─ app.py
```

```
|─ config/
```

```
|─ database/
```

```
|─ data/
```

```
|   │─ raw/
```

```
|   │─ processed/
```

```
|   └─ archive/
```

```
|
```

```
|─ etl/
```

```
|─ analytics/
```

```
|─ ai/
```

```
├─ dashboard/
├─ reports/
├─ alerts/
├─ scheduler/
├─ exports/
├─ assets/
├─ logs/
├─ tests/
└─ docs/
```

11. User Flow

Launch Application

↓

Import Blinkit Data

↓

Data Processing

↓

Dashboard Generation

↓

Business Analysis

↓

AI Recommendations

↓

Executive Decisions

12. Dashboard Navigation

- Executive Dashboard

- Sales Intelligence
 - Regional Intelligence
 - Keyword Intelligence
 - Competition Intelligence
 - Warehouse Intelligence
 - Smart Alerts
 - AI Business Advisor
 - Reports
 - Settings
-

13. Security

- Local database encryption (future)
 - Role-based authentication (future)
 - Audit logging
 - Automatic database backups
 - Secure API key management
 - Input validation
 - Error logging
-

14. Success Metrics

Business KPIs:

- Increased sales growth
- Improved inventory turnover
- Reduced stock-outs
- Improved keyword conversions
- Faster decision making
- Reduced manual reporting effort

Technical KPIs:

- Report generation under 60 seconds
 - Dashboard load under 3 seconds
 - Daily automated processing
 - Modular codebase
 - High test coverage
 - Reliable scheduled execution
-

15. Development Roadmap

Phase 1

- Project setup
- Database design
- Blinkit data import
- Executive Dashboard
- Sales Intelligence

Phase 2

- Regional Intelligence
- Warehouse Intelligence
- Smart Alerts
- Automated reporting

Phase 3

- Keyword Intelligence
- Competition Intelligence
- AI Business Advisor

Phase 4

- Predictive analytics
 - Demand forecasting
 - Advanced reporting
 - Multi-marketplace support
-

16. Guiding Principle

RealNut Intelligence is not a reporting tool.

It is a business decision support system.

Every page, chart, report, and recommendation must answer one question:

"What action should Real Nut take today to improve revenue, profitability, inventory efficiency, or market position?"

If a feature cannot drive a business decision, it should be reconsidered before implementation.